Jasper Sands

Palo Alto, CA | (650) 924-8429 | jaspersands02@gmail.com | js6908@columbia.edu
jaspersands.com | github.com/Jaspersands | linkedin.com/in/jaspersands

RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for qubit characterization and control · simulation of quantum systems on near-term hardware.

EDUCATION

Columbia University, New York, NY — Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Machine-Assisted Mathematics · Neural Networks & Deep Learning · Machine Learning for Functional Genomics.

Washington University in St. Louis, St. Louis, MO — Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

Coursework: Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison · Parallel & Concurrent Programming · Computer Security · Reverse Engineering & Malware Analysis · Econometrics · Game Theory.

DIS Copenhagen, Copenhagen, Denmark · *Study Abroad* — Spring 2024

Coursework: Artificial Neural Networks and Deep Learning · Artificial Intelligence.

TECHNICAL REPORTS

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* Columbia University, 2025. jaspersands.com/assets/practical-llm-based-lossless-compression.pdf

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* — Sep 2026 – May 2027

- Automating silicon spin qubit characterization under Dr. Nard Dumoulin Stuyck, targeting the manual tuning and interpretation of charge measurements that currently bottleneck semiconductor quantum hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Evaluating the automated analysis against how an expert reads the same device by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* — Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry.
- Tuned for precision against a fixed false-positive budget on live traffic, since a false positive quarantines real mail.

Desdr Open Insurance Toolkit, Columbia University · *Graduate Researcher* — Sep 2025 – Dec 2025

- Extended an open-source insurance modeling toolkit with Dr. Eugene Wu, built for NGOs serving rural farmers.
- Turned SMS survey data into actuarial and climate-risk features under severe data sparsity, and built models estimating expected agricultural losses to support micro-insurance program design.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* — May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian by grounding outputs in ontology and molecular graph data.
- Built tooling turning natural-language drug queries into graph-derived checks on AlphaFold structures.

Foster System Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents.
- Implemented a human-in-the-loop feedback workflow deployed on Hugging Face Spaces, enabling cross-state policy comparison and interactive analysis of foster-care outcomes.

Computer Vision & AI Lab, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – Jun 2025

- Built distributed collection and processing pipelines under Dr. Umar Iqbal that parsed billions of forum posts for population-scale behavioral and privacy research.
- Applied statistical modeling to demographic and behavioral patterns in online discourse.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation for the State Street and Classiq challenge.
- Measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ quasi-Monte Carlo classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes with real-time error injection and syndrome visualization, running exact minimum-weight perfect matching and Union-Find decoding entirely client-side.
- Finite-size scaling estimates thresholds of 3.2% with MWPM, 2.7% with Union-Find, 14.7% at code capacity.

Q-Search: proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian hidden subgroup, dihedral coset, and linear code equivalence problems, pairing frontier theoretical reasoning with agentic engineering.
- Formal proof gates test each proposed mechanism against 1,200+ classical baselines across 720+ theorem modules, with negative results permanently indexed so the engine does not re-propose a dead end.

Noise-Aware Adiabatic Simulation: transverse-field Ising model · *Qiskit* IBM Heron hardware

- Digitized adiabatic ground-state preparation on Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps for second-order accuracy.
- Flattened the anneal schedule at the critical point, where a linear sweep races through the closing gap.

LLM-Based Lossless Text Compression · *PyTorch, CUDA* jaspersands.com/llmcompression

- Arithmetic coding driven by autoregressive language model predictions across five models on the Canterbury Corpus, reaching 5–16× average and 39× best-case compression against 2–4× for gzip and zstd.
- Static KV caching, CUDA graph capture, and batched segment processing made throughput measurable; the pipeline still runs 100–1000× slower than the classical compressors it beats.

Entanglement and Matrix Product States: many-body systems slides

- Sequential SVD wavefunction decomposition measuring Schmidt spectra, entanglement entropy at every cut, and fidelity under truncation, relating bipartite entanglement structure to MPS compressibility.

Other work: QSVT spectrum transformations, reshaping a matrix spectrum with quantum signal processing (slides) · HyperAMR, embedding metagenomic contigs and resistance annotations in one hyperbolic space (write-up).

TEACHING EXPERIENCE

Introduction to Quantum Computing, Columbia University · *TA* Jan 2026 – May 2026

- Designed problem sets on quantum circuit simulation and algorithm analysis, and supported students on measurement, entanglement, and the standard algorithm set.

Parallel and Concurrent Programming, Washington University in St. Louis · *TA* Jan 2025 – May 2025

- Designed and evaluated assignments on concurrency, synchronization, and performance analysis.

Reverse Engineering and Malware Analysis, Washington University in St. Louis · *TA* Aug 2024 – Dec 2024

- Supported an upper-level systems and security course and led review sessions before projects and exams.

INDUSTRY EXPERIENCE

Smack Technologies · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulation for the core pipelines, integrating the knowledge graph to expand corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency, increasing corpus-generation throughput 64×.
- Validated simulator outputs against independently selected reference cases, bounding error to 0.55%, in an automated report that regenerates as the physics model changes.

Highnote · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day.
- Wrote 100+ rule-based and ML anomaly detections for suspicious authentication, command execution, and data access.

Mindtrip · *Software Engineer Intern* May 2023 – Jul 2023

- Built internal admin and secure-query tooling in JavaScript and Ruby on Rails through Retool, replacing direct production database writes and saving roughly 250 engineering hours a month.

Bugcrowd · *Associate Application Security Engineer* May 2020 – Jul 2020

- Triaged and validated vulnerability submissions from a global researcher community, working with enterprise clients on severity prioritization and disclosure.

AWARDS AND COMPETITIONS

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street and Classiq challenge.
- **Qualified**, NYU Abu Dhabi Quantum Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

TECHNICAL SKILLS

Quantum software: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution and benchmarking.

Quantum methods: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT and block encoding, linear combination of unitaries, quantum walks, VQE and QAOA, Trotter decomposition.

Error correction: Rotated and XZZX surface codes, stabilizer formalism, minimum-weight perfect matching and Union-Find decoding, finite-size scaling and threshold estimation, randomized benchmarking, tomography, noise modeling.

Theory: Quantum complexity (BQP, QMA), oracle and query complexity, asymptotic analysis, linear algebra and numerical methods, tensor networks, probability, convex and stochastic optimization.

Programming: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java.

Machine learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy, Pandas, NetworkX.

Systems and tooling: Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, Jupyter, LaTeX.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025 – Jun 2026). Ran weekly technical sessions covering all 33 chapters of Andrew Childs' quantum algorithms lecture notes, each paired with simulation checks, and brought in faculty and PhD students to speak.
- **Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.
- **Team Captain**, WashU Gymnastics Club.
- **Student Organizer**, WashU Votes. Ran voter registration drives and lobbied at the Missouri State Capitol.